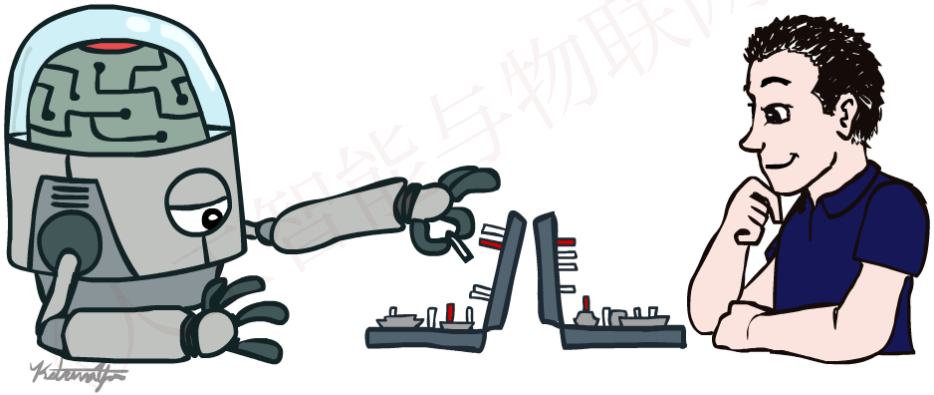
Artificial Intelligence & IoT

Introduction



Instructor: Kaikai Pan

Zhejiang University

[These slides were adapted from the slides by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. http://ai.berkeley.edu.]

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Course Staff

Instructor

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TA

徐震

Zhen Xu

USSLAB（智能系统安全）

Slogan: Looking for troubles

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Course Information

- Communication:
  - My email:
    - kaikaipan15@gmail.com
    - pankaikai@zju.edu.cn
  - TA email:
    - xu_zhen_yx@163.com
  - Wechat group (**微信群**)
  - DingTalk group
  - Learning in ZJU (学在浙大)
- Work and Grading:
  - Three homework assignments: 36%
  - Final exam: 58%
  - Attendance: 6%
  - Slides in English
  - Lecture in Chinese

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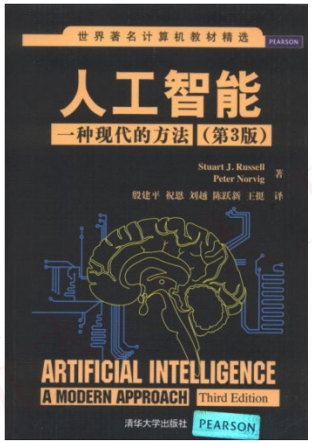
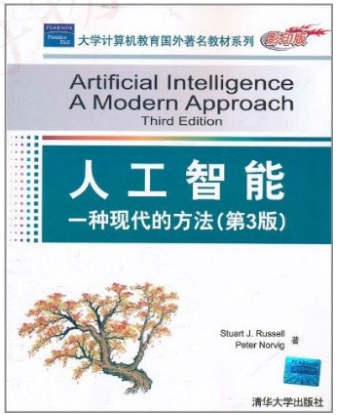
Reference Book

- Not required, but for students who want to read more we recommend
  - Russell & Norvig, AI: A Modern Approach, 3rd Ed.
  - 人工智能-一种现代化的方法(第三版)

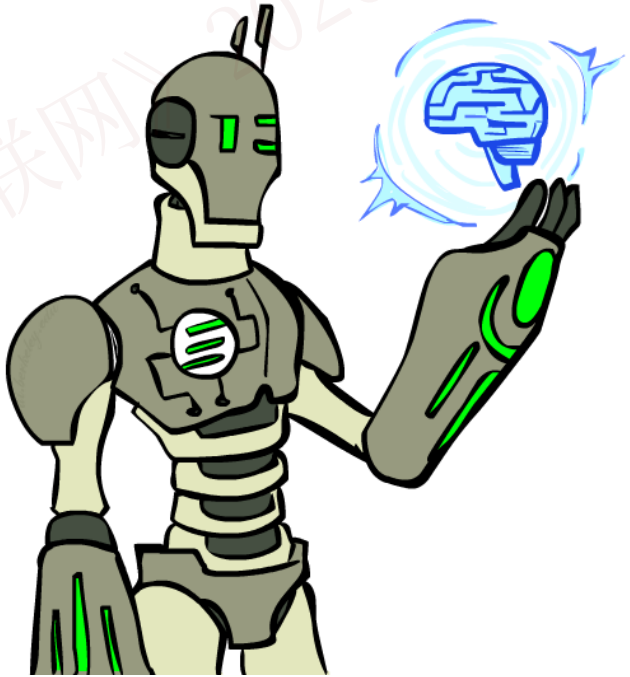




  - Warning: Not a course textbook. The slides do not necessarily follow the presentation in the book.

Today

- What is artificial intelligence?
- What can AI do?
- What is this course?



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What is AI?

The science of making machines that:

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Rational Decisions

- We'll use the term **rational (理性)** in a very specific, technical way:
  - Rational: maximally achieving pre-defined goals (目标)
  - Rationality only concerns what decisions (决策) are made (not the thought process behind them)
  - Goals are expressed in terms of the utility (效用) of outcomes
  - Being rational means **maximizing your expected utility**

A better title for this course would be:

Computational Rationality

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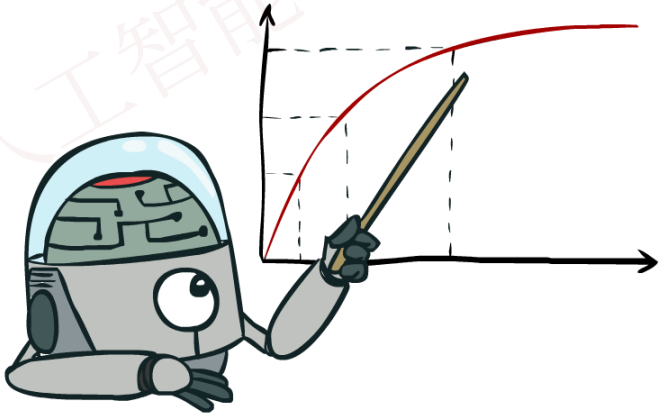
Maximize Your Expected Utility



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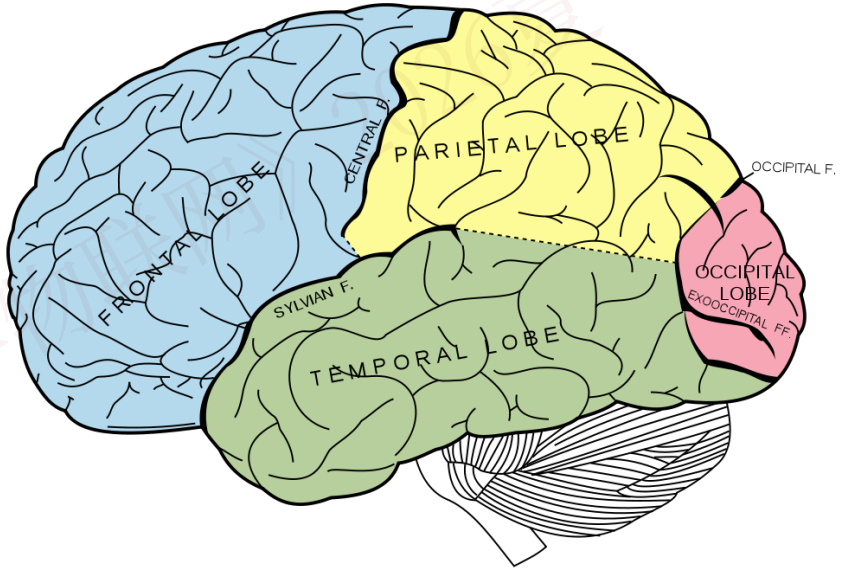
What About the Brain?

- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- "Brains are to intelligence as wings are to flight"
- Lessons learned from the brain:
  - Problem solving
  - Reasoning
  - Learning



Course Outline

- Part I: Solving Problems
  - Search
  - Game
- Part II: Reasoning
  - Probabilistic reasoning
  - Decision network
- Part III: Learning
  - Machine learning
  - Reinforcement learning
- Part IV: Advanced Materials (Tentative)
  - Embodied AI
  - AI for EE…

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Three Schools of AI

- Symbolism (符号主义)
- Connectionism (联结主义)
- Behaviorism (行为主义)

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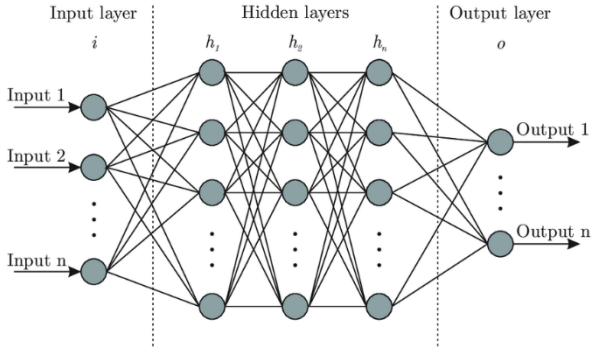
Three Schools of AI

- Symbolism (符号主义)
  - a.k.a, rule-based AI, classic AI and good old-fashioned AI (GOFA)
  - Intuition: Humans use symbols and their relationships for cognition
    - A symbol is a pattern that stands for other things.
    - Syllogism: All men are mortal; Aristotle is a man; therefore Aristotle is mortal.
  - Symbolic AI: Use knowledge in reasoning and learning (with only modest input data)
  - Example: expert system, if/then pairing to instruct how to behave
  - Applications with clear-cut rules (√)
  - Applications with variations and uncertainty (×)

Three Schools of AI

- Connectionism (联结主义)
  - Intuition: Human brain works with interconnected neurons
  - Connectionist AI: Learn from data with little or no prior knowledge
  - Example: Artificial neural network
  - Applications with massive high-quality training data
  - Hard to explain (maybe…)



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Three Schools of AI

- Behaviorism (行为主义)
  - Intuition: Humans respond to stimulus, i.e., punishment and reward
  - Behaviorist AI: Use statistical learning to better mine and predict data
  - Example: Reinforcement learning
  - Applications that learn from experience, e.g., robot
  - Markovian assumption of events. The curse of dimensionality.

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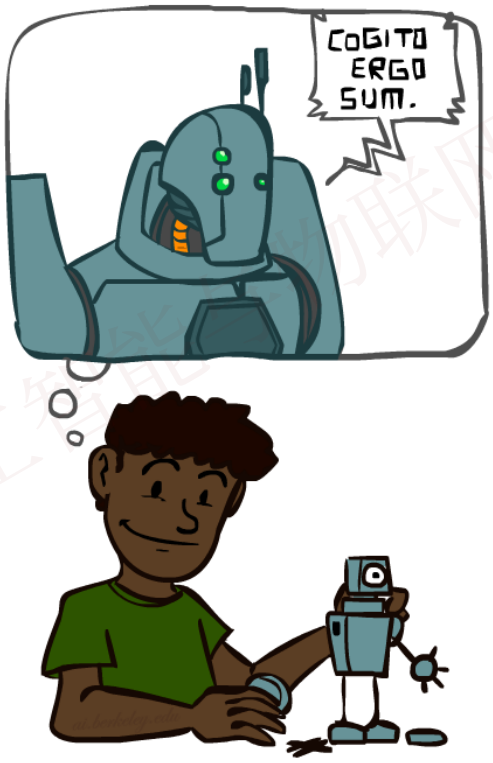
A (Short) History of AI



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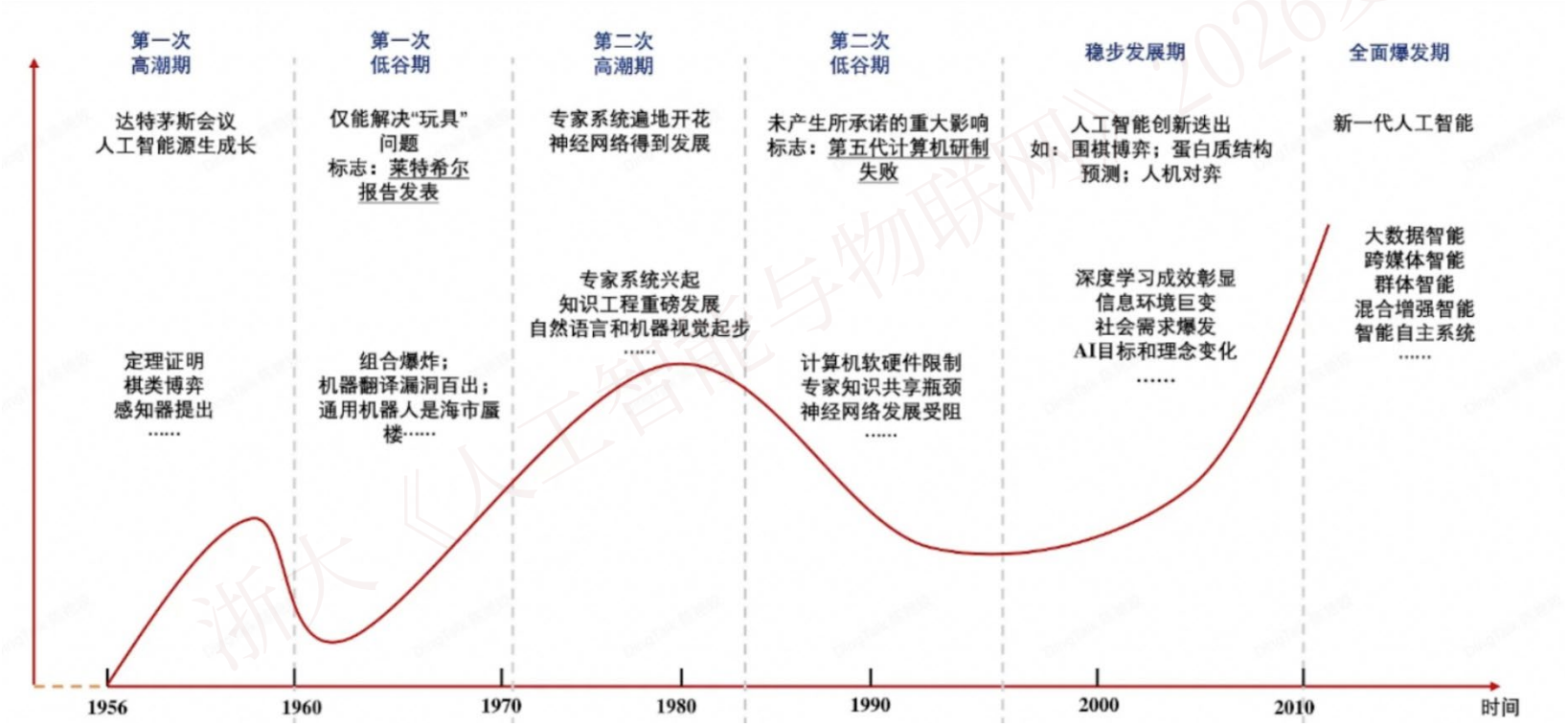
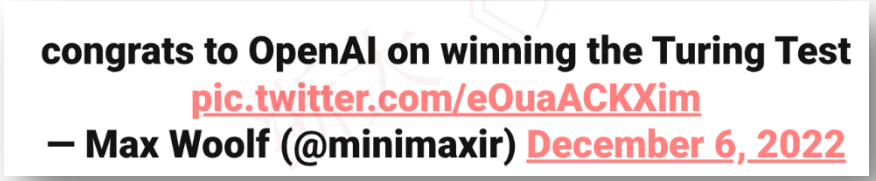
A (Short) History of AI



First Golden Age

- Alan Turing is often seen as the father of Artificial Intelligence.
- He invented the Turing Test, designed to determine if a computer system can be called an artificial intelligence or not, based on whether it can fool a human into thinking it is human too.
- No system has yet passed the Turing Test.
  - Has ChatGPT passed the Turning Test?
  - Some said yes, but some said no…



1956 Dartmouth Conference: Proposal

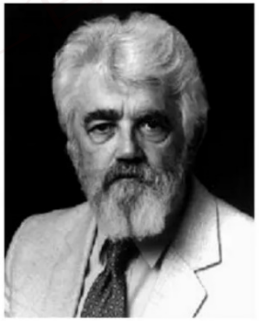
"We propose that a 2 month, 10 man study of **artificial intelligence** be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that ***every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it***. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves."

---- J. McCarthy (Dartmouth College)
M. L. Minsky (Harvard University)
N. Rochester (I.B.M. Corporation)
C.E. Shannon (Bell Telephone Laboratories)

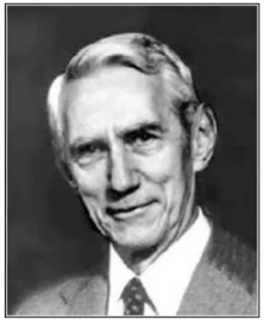
1956 Dartmouth Conference: The Founding Fathers of AI

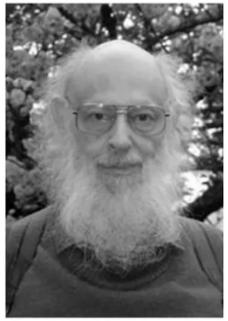
- 1965

- 2000+


John McCarthy

Marvin Minsky


Claude Shannon


Ray Solomonoff

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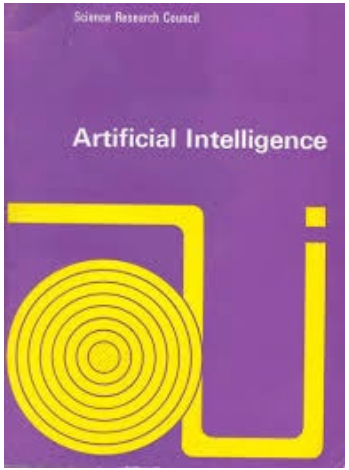
1956 Dartmouth Conference: Topics

Topic	Description
Automatic Computers	Build an automatic calculator to simulate the machine
How Can a Computer be Programmed to Use a Language	Human thought consists of manipulating words for reasoning. Formulate the machine to use language.
Neuron Nets	How can a set of neurons be arranged so as to form concepts.
Theory of the Size of a Calculation	A method of measuring the complexity of calculating devices.
Self-improvement	Intelligent machine will carry out activities for self-improvement.
Abstractions	Form abstractions from sensory and other data.
Randomness and Creativity	Difference between creative thinking and unimaginative competent thinking.

1973 Lighthill Report



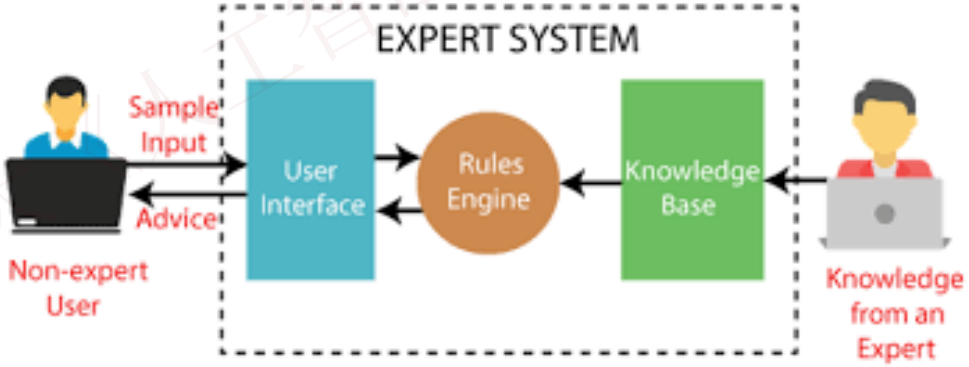
- Lighthill's report was commissioned by the Science Research Council (SRC) to give an unbiased view of the state of AI research primarily in the UK.
- Lighthill's report provoked a massive loss of confidence in AI. It persisted for almost a decade.
- **Sir Michael James Lighthill** was a British Mathematician.

“Most workers in AI research and in related fields confess to a pronounced feeling of disappointment in what has been achieved in the past twenty-five years. ”

----Lighthill Report: Artificial Intelligence: A General Survey

1972 MYCIN: Expert System

- MYCIN was an early expert system to diagnose and recommend treatment for blood infections.
  - Developed at Stanford University.
  - About 600 IF-THEN rules.
  - Research indicated that MYCIN had an accuracy of ~65%, better than average human experts.



1982-1994 Fifth Generation Computer Systems

- A 10-year initiative launched by Japan. The aim was to build “epoch-making” computers for artificial intelligence applications.
- Reliance on concurrent logic programming
  - Concurrent logic programming was eventually surpassed by technologies like Sun workstations and Intel x86 machines.
  - Concurrent logic programming, specifically Prolog, wasn't widely adopted in the mainstream computer.

- "Ten years ago we faced criticism of being too reckless," said Kazuhiro Fuchi, Fifth Generation's head. "Now we see criticism from inside and outside the country because we have failed to achieve such grand goals.”

---- *New York Times*

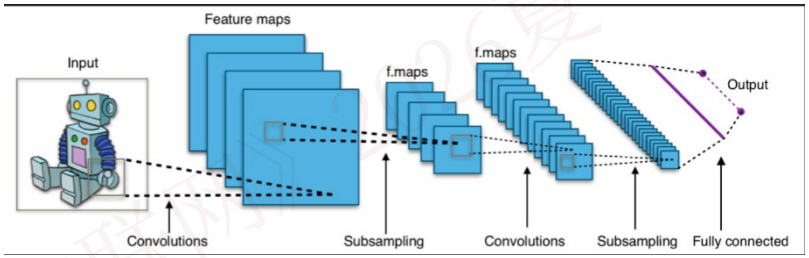
Quiet Progress

- On May 11, 1997, Deep Blue beat Kasparov.

- 2004-present, DARPA Grand Challenge

Quiet Progress

- In 2011, the IBM Waston computer system competed on *Jeopardy!* against champions B. Rutter & K. Jennings, winning $1 million.



- 1998, Yann LeCun, Convolutional Neural Networks.

- 2012, Geoffrey E. Hinton, ImageNet Classification with Deep Convolutional Neural Networks

Second Gold Age

- 2012: Hinton uses Deep ConvNets to solve ImageNet classification
- 2013: Hinton hired by Google, uses NN on mobile
  - Boston Dynamics acquired by Google
  - Yann LeCun heads FAIR for Facebook
- 2014: Google acquires DeepMind
- 2015: Musk donates $10m to Future of Life to study existential risk for artificial intelligence
  - Autonomous vehicle testing begins
- 2016: AlphaGo beats Lee Sedol
- 2017: AlphaZero beats Stockfish

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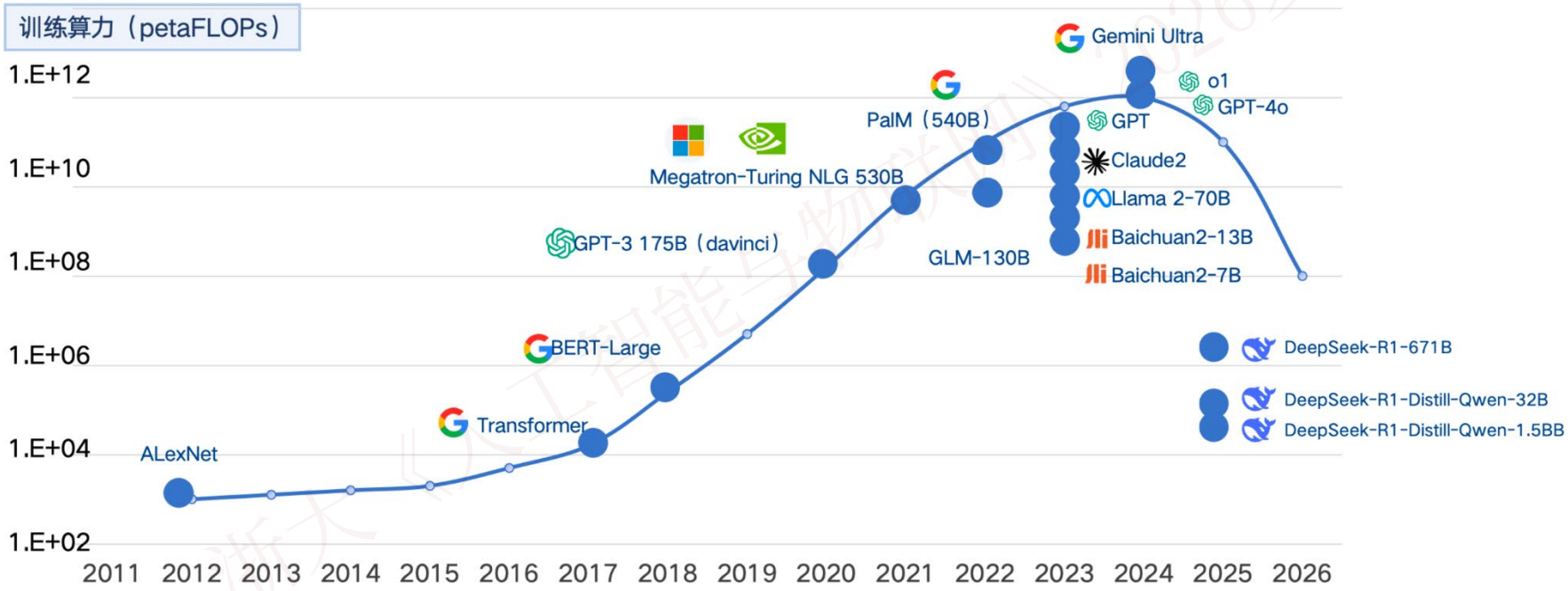
2020--: Large Language Model



What Can AI Do?

- DeepSeek: Large Language Model
- Manus: AI Agent
- Unitree: Robot

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DeepSeek

https://sider.ai/chat

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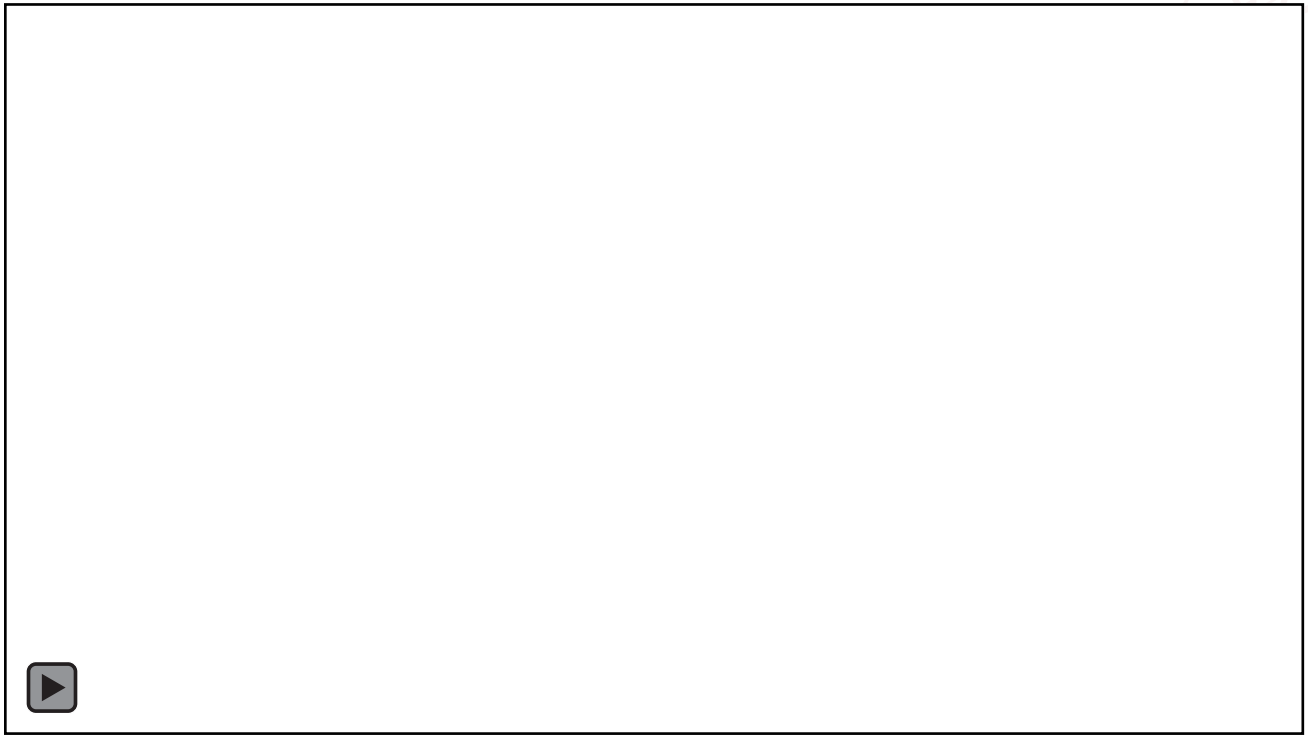
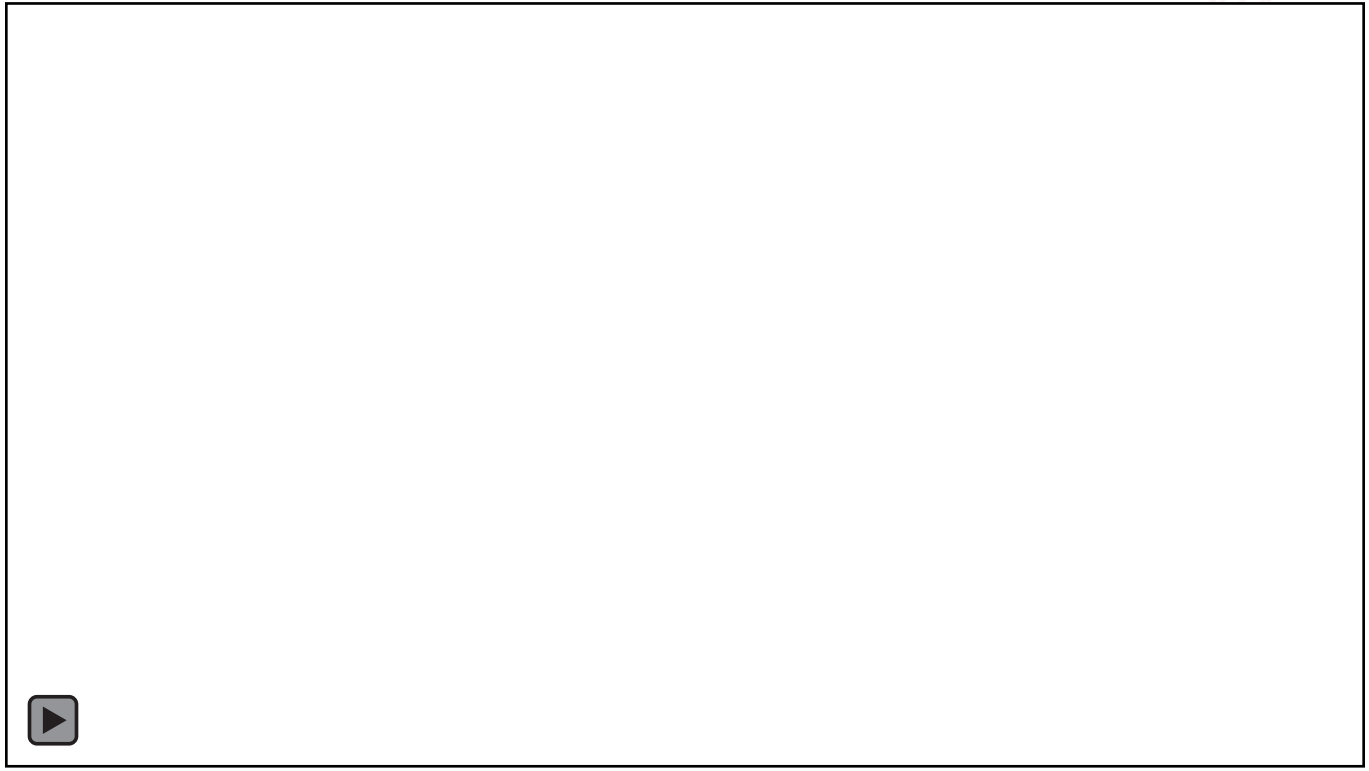
Manus AI Agent

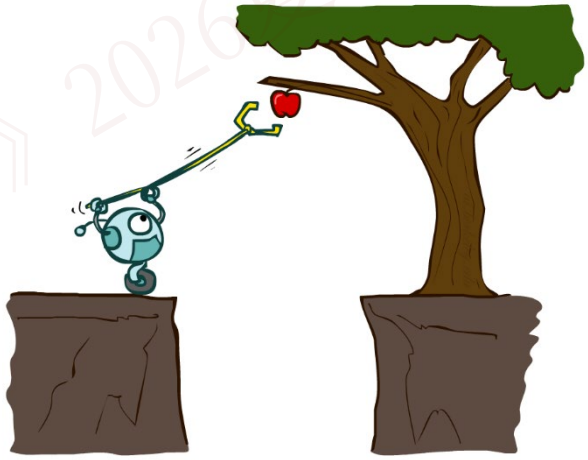


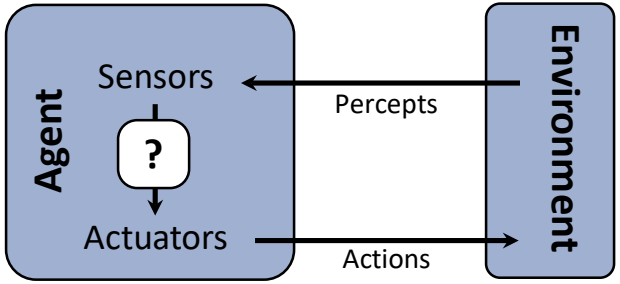
Unitree Robot



What is this course?

- Designing rational agents
  - An **agent (智能体)** is an entity that perceives and acts.
  - A **rational agent** selects actions that maximize its (expected) **utility**. (uncertainty)
- Characteristics of **percepts (感知输入)**, **environment (环境)**, and **action space (行动空间)** determine how to select rational actions
- **This course** is about:
  - General AI techniques for a variety of problems
  - Learn to recognize *when & how a new problem* can be solved with existing techniques (**方法论**)





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